

DSA210 Term Project: TV Series Analysis

Final Report

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1. Overview & Motivation

The main goal of this project is to investigate the relationship between the quality of a TV series and its mainstream popularity. In the entertainment industry, it is often assumed that high-quality content automatically attracts a larger audience. I wanted to test this assumption using real data by analyzing whether a higher IMDb rating correlates with a larger audience size (number of votes). Additionally, I wanted to see if older, classic TV series hold higher quality standards compared to modern ones.

2. Data Collection & Preprocessing

I created a dataset containing more than 250 popular TV series. The data was gathered using Python, extracting key variables from IMDb:

- `averageRating`: The numeric score representing the quality of the show.
- `numVotes`: The total number of user votes, representing the audience size.
- `startYear`: The release year of the show to analyze time-based trends.

During the data enrichment phase, I also explored Google Trends data using an API. I discovered that the API returned a constant peak value of 100 for all series rather than a moving average. Instead of discarding this, I kept it as a key Exploratory Data Analysis (EDA) finding regarding data normalization in real-world APIs, which helped me focus deeper on the solid IMDb dataset variables.

3. Exploratory Data Analysis (EDA)

To understand the distribution of the data, I generated several plots to examine our main features:

- **IMDb Ratings:** The ratings show a nearly normal distribution centered around 7.5 to 8.5, indicating a high baseline quality for the selected series.
- **Audience Size (numVotes):** The distribution is heavily right-skewed, meaning that while most shows have a moderate, steady audience, a few mega-hits pull in millions of votes as extreme outliers.

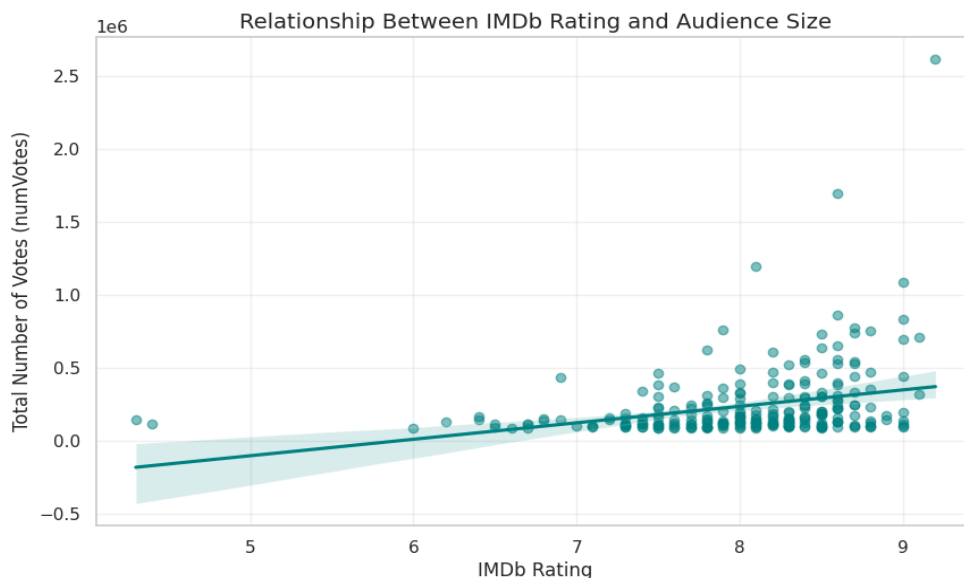


Figure 1: Scatter plot showing the weak linear relationship between IMDb ratings and total user votes.

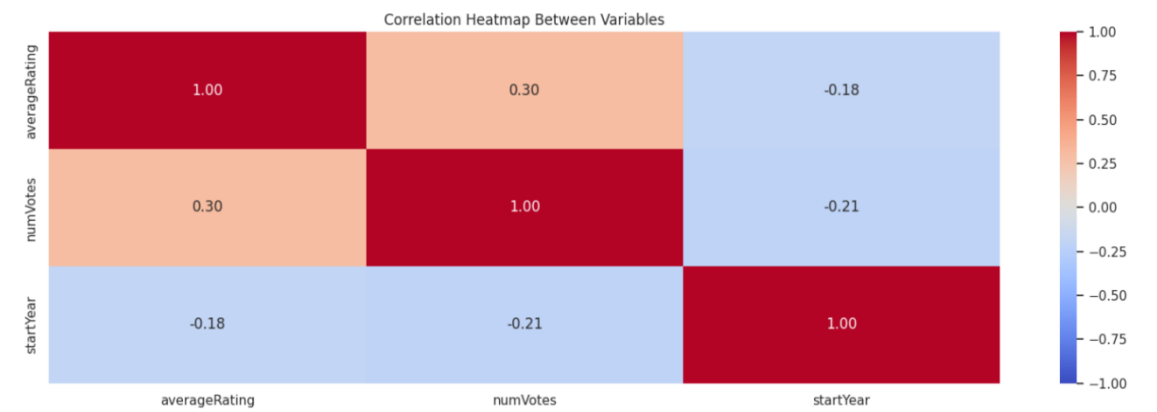
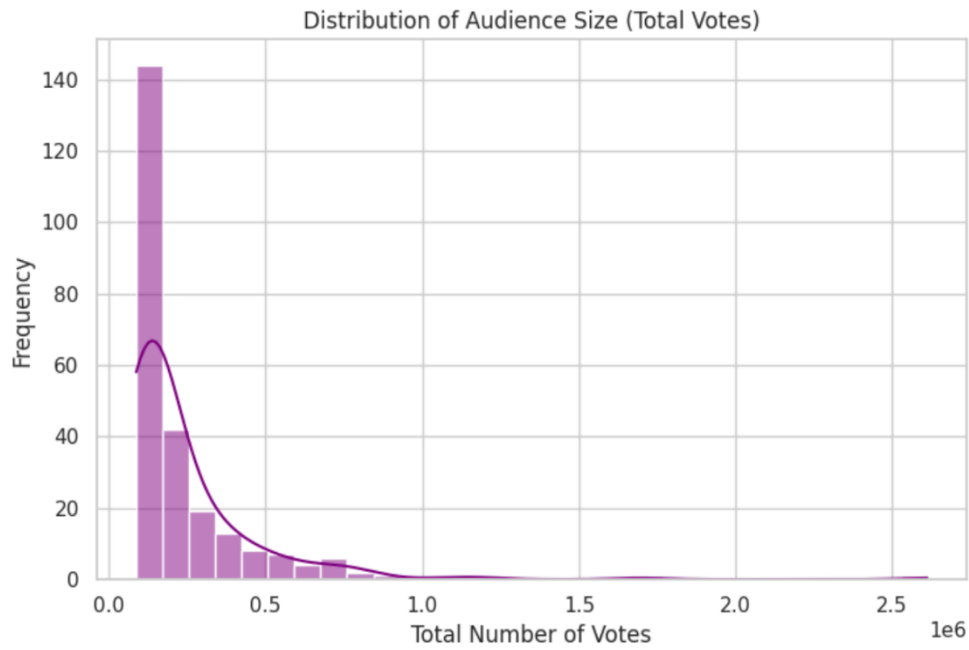


Figure 2; Distribution histograms of ratings and votes alongside the overall correlation heatmap.

4. Hypothesis Testing

To validate my visual findings with statistical evidence, I conducted two main hypothesis tests:

Test 1: Rating vs. Audience Size (Pearson Correlation)

- **Output:** Pearson Correlation: 0.30 (p-value: 0.00000)
- **Analysis:** The test reveals a statistically significant but weak positive correlation. This confirms that higher quality does help drive popularity, but a high rating alone does not guarantee a massive mainstream audience.

Test 2: Classic Series vs. Modern Series (Independent T-Test)

- **Output:** T-Test Statistic: 2.809 (p-value: 0.00539)
 - *Average Rating of Older Shows (< 2017): 8.17*
 - *Average Rating of Newer Shows (>= 2017): 7.94*
- **Analysis:** With a p-value well below 0.05, we reject the null hypothesis. There is a statistically significant difference proving that older shows in this dataset have a higher average quality score than newer releases.

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HYPOTHESIS TEST 1: Rating vs. Audience Size
Pearson Correlation: 0.30 (p-value: 0.00000)
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Interpretation: Weak positive correlation. A high rating does not guarantee mass-market popularity, but they are somewhat aligned.

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HYPOTHESIS TEST 2: Do newer shows get different ratings than older ones?
Independent T-Test Statistic: 2.809 (p-value: 0.00539)
Average Rating of Older Shows (< 2017): 8.17
Average Rating of Newer Shows (>= 2017): 7.94
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5. Machine Learning (K-Means Clustering)

To uncover hidden structures in the TV series ecosystem, I applied the **K-Means Clustering** algorithm.

- **Optimal K Selection:** Using the **Elbow Method**, I plotted the Sum of Squared Errors (SSE) against different cluster sizes. The curve showed a distinct "elbow" point at $k=4$, which was selected as the optimal number of groups.

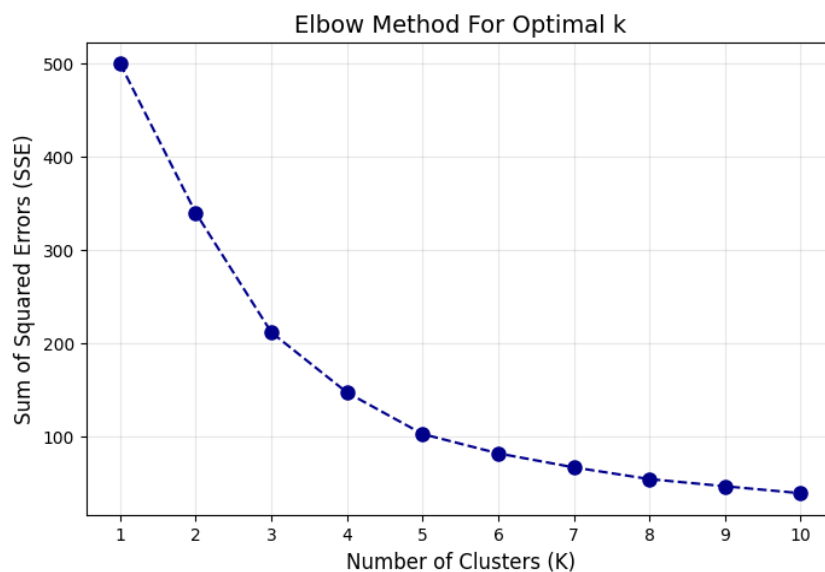


Figure 4: Elbow method metric plot used to determine the optimal number of clusters ($k=4$).

By feeding the scaled features into the model, the algorithm successfully divided the TV series into 4 distinct real-world profiles without any human labeling:

1. **Cluster 3 (The Mega-Phenomena):** Shows with both exceptional quality scores and massive, record-breaking vote counts (e.g., *Game of Thrones*, *Stranger Things*).
2. **Cluster 0 (Mainstream Masterpieces):** High-quality, widely watched series that successfully captured around 1 million votes (e.g., *The Boys*, *Better Call Saul*, *Sherlock*).
3. **Cluster 2 (Hidden Gems):** Critically acclaimed productions with outstanding ratings but smaller, more specific audiences between 300k and 400k votes (e.g., *Succession*, *Severance*, *Mindhunter*).

4. **Cluster 1 (Popcorn Entertainment):** Shows that achieved high popularity and large audience sizes but received average or lower critical ratings between 6.9 and 7.5 (e.g., *The Flash*, *13 Reasons Why*, *The Rings of Power*).

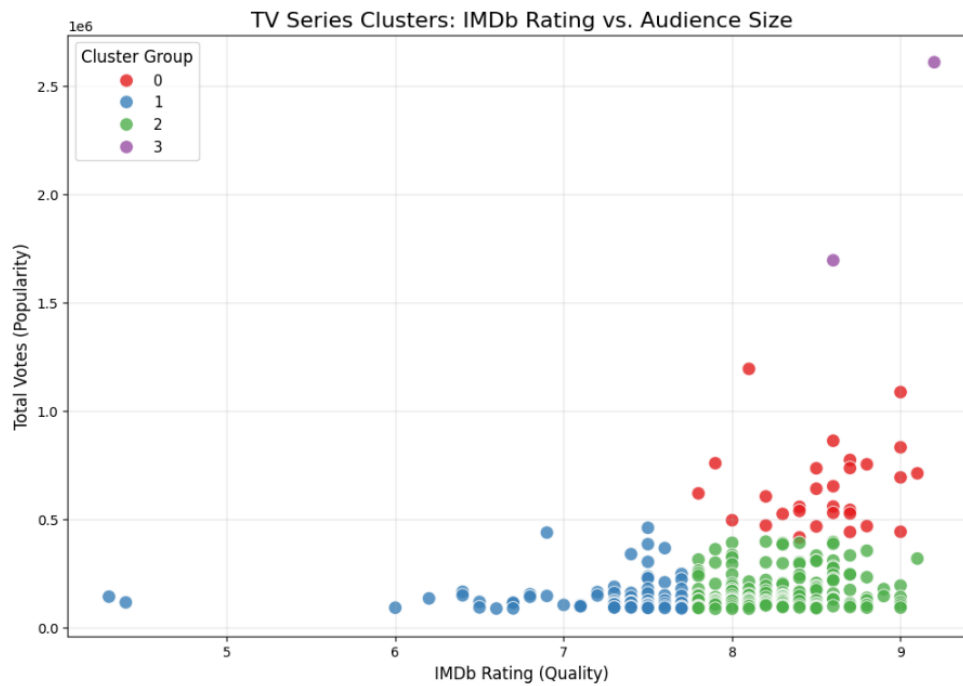


Figure 5: Scatter plot representing the final 4-cluster model output using different colors.

--- CLUster 0 SAMPLES ---

primaryTitle	averageRating	numVotes
The Walking Dead	8.1	1194752
Sherlock	9.0	1087604
The Boys	8.6	862839
Better Call Saul	9.0	832588
Peaky Blinders	8.7	774225
Squid Game	7.9	759230
True Detective	8.8	754142
Black Mirror	8.7	736658
The Last of Us	8.5	735572
Attack on Titan	9.1	712492

--- CLUster 1 SAMPLES ---

primaryTitle	averageRating	numVotes
Arrow	7.5	460790
The Lord of the Rings: The Rings of Power	6.9	438848
The Flash	7.5	385288
You	7.6	366883
13 Reasons Why	7.4	339286
The 100	7.5	303611
Once Upon a Time	7.7	247739
Agents of S.H.I.E.L.D.	7.5	235143
Bridgerton	7.5	227792
Monster	7.7	222767

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--- CLUster 2 SAMPLES ---
primaryTitle averageRating numVotes
Sex Education      8.2    397371
Mindhunter         8.6    395935
Lucifer            8.0    391796
Ozark              8.4    391648
Fallout            8.3    390751
Severance          8.6    387326
Homeland           8.3    384236
American Horror Story 7.9    361936
Succession         8.8    355191
Orange Is the New Black 8.0    336152
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--- CLUster 3 SAMPLES ---
primaryTitle averageRating numVotes
Game of Thrones      9.2   2611033
Stranger Things       8.6   1696143
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Figure 6: Python output listing the top TV series classified under each cluster group.

6. Conclusion & Personal Observations

This project successfully demonstrated how unsupervised machine learning and statistical testing apply to the entertainment industry. The most fascinating finding was how the K-Means algorithm naturally discovered the "Hidden Gems" and "Popcorn Entertainment" categories purely from raw numerical data.

From my perspective, When I first started, I honestly believed it was a simple rule: if a TV show is a masterpiece, it will automatically get millions of viewers. Seeing the math prove me wrong was really surprising. This project taught me the critical importance of data preprocessing, feature scaling and how to properly handle real-world API anomalies. Interpreting statistical metrics like p-values allowed me to back up everyday cultural observations with mathematical proof.

7. References

- IMDb Datasets (Non-Commercial Licensing Metadata)
- Google Trends API (Used for analyzing historical search interest and momentum)
- Python Libraries: pandas, numpy, seaborn, matplotlib, scikit-learn, scipy.stats
- AI Assistance: Gemini (Large Language Model by Google) – Used as an AI assistant for data storytelling, code optimization, structural organization, and formatting of the final report.

<https://kamilal-hub.github.io/DSA210-TERM-PROJECT/#overview>
my website

